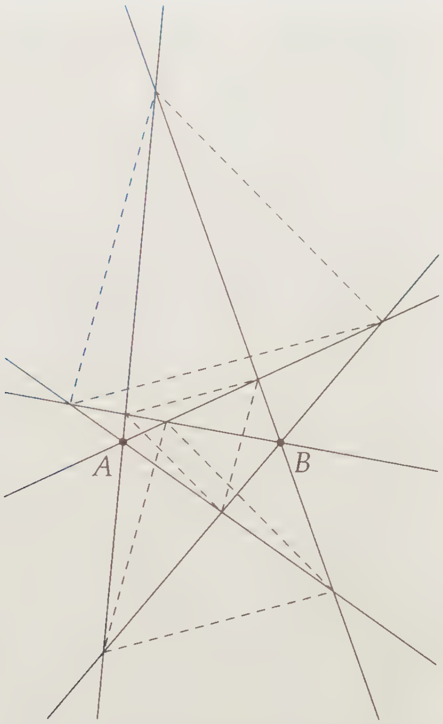


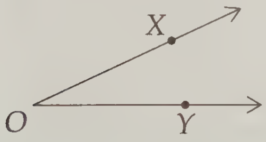
The Lighthouse Theorem

We're going to turn this team around 360 degrees. – Jason Kidd

CHAPTER 2 Angles

2.1 What is an Angle?

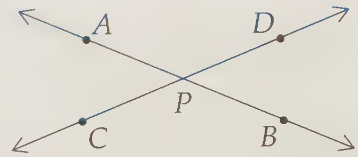
When two rays share an origin, they form an **angle**.



In the diagram at left, rays $\overrightarrow{OX}$ and $\overrightarrow{OY}$ share origin O. We can refer to the angle they form as $\angle XOY$. The $\angle$ symbol tells us we're referring to an angle. The common origin is called the **vertex** of the angle, and the rays $\overrightarrow{OX}$ and $\overrightarrow{OY}$ are called the **sides** of the angle. Notice that when we write the angle as $\angle XOY$, we put the vertex in the middle. We could also refer to the angle as $\angle YOX$, but not as $\angle XYO$. When it's very clear what angle we're talking about, we can just name it with the vertex: $\angle O$.

Of course, two intersecting lines also make angles.

Lines $\overleftrightarrow{AB}$ and $\overleftrightarrow{CD}$ at right intersect at P. Here, we can't just write $\angle P$, since there are many different possible angles this could mean, such as $\angle APC$, $\angle APD$, $\angle DPB$, or $\angle BPC$. We might even be referring to $\angle APB$.



Now that we know what angles are, we need a way to measure them so we can compare one angle to another.

2.2 Measuring Angles

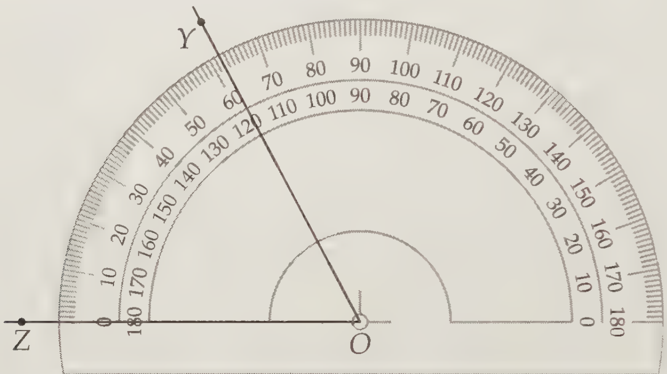


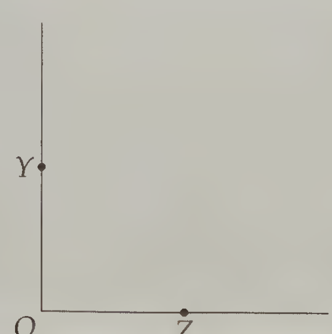
Figure 2.1: A Protractor

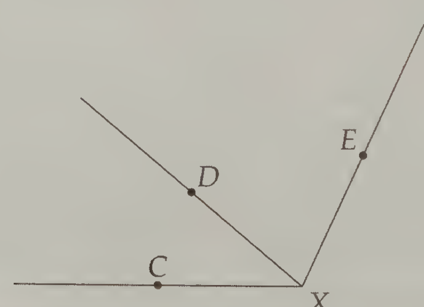
Just as we use a ruler to measure the lengths of segments, we can use a **protractor** to measure angles. Roughly speaking, an angle's measure is how 'open' the angle is. Our protractor above shows half a circle (which we call a **semicircle**) divided into 180 equal pieces. Each of these little pieces is considered one **degree** of the semicircle, so that an entire circle is 360 degrees. We use the symbol $^\circ$ to denote degrees, so that a whole circle is 360°.

We use a protractor to measure the number of degrees of a circle between the two sides of an angle whose vertex is the center of the circle. For example, in Figure 2.1, the vertex of $\angle YOZ$ is placed at the center of the semicircle. There are 62 degrees between sides $\overrightarrow{OZ}$ and $\overrightarrow{OY}$ of $\angle YOZ$, so we say that $\angle YOZ = 62^\circ$. Sometimes angles are written with an m before $\angle$ to indicate measure: $m\angle YOZ = 62^\circ$.

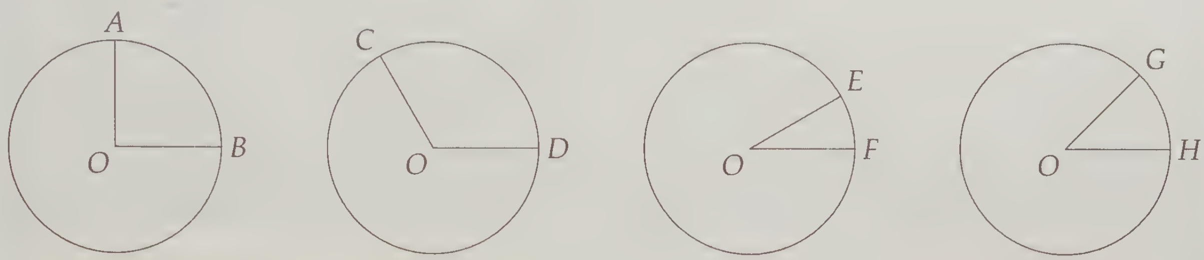
Problems

Problem 2.1: Use your protractor to find $\angle YOZ$, $\angle CXD$, $\angle DXE$, and $\angle CXE$.





Problem 2.2: The diagram below shows four common angles. In each case, point O is the center of the circle. $\angle AOB$ cuts off 1/4 of a circle, $\angle COD$ cuts off 1/3 of a circle, $\angle EOF$ cuts off 1/12 of a circle, and $\angle GOH$ cuts off 1/8 of a circle.



(a) What is the measure in degrees of $\angle AOB$?

(b) What is the measure in degrees of $\angle COD$?

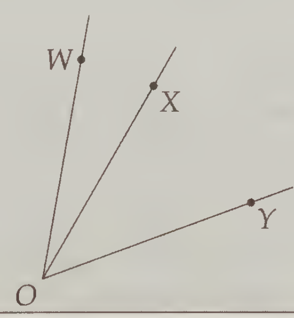
(c) What is the measure in degrees of $\angle EOF$?

(d) What is the measure in degrees of $\angle GOH$?

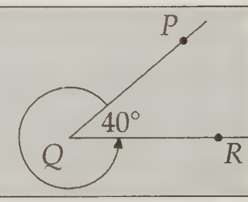
(e) What's so special about 360; why do we use 360 for the number of degrees in a whole circle?

Do not use a protractor; use what you are told about the angles in the text.

Problem 2.3: Given that $\angle WOY = 60°$ and $\angle WOX = 20°$ below, find $\angle XOY$.



Problem 2.4: Suppose instead of measuring an angle the 'regular' way, we go the 'long' way around, as shown in the diagram. The 'regular' angle PQR has measure 40°. What is the measure of the 'long' way around angle?

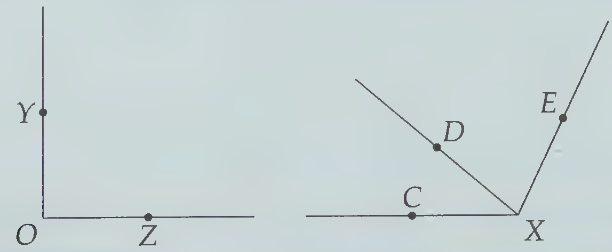


Problem 2.5: Use your protractor to create a 37° angle and a 143° angle.

Extra! *We could use two Eternities in learning all that is to be learned about our own world and the thousands of nations that have arisen and flourished and vanished from it. Mathematics alone would occupy me eight million years.*

–Mark Twain

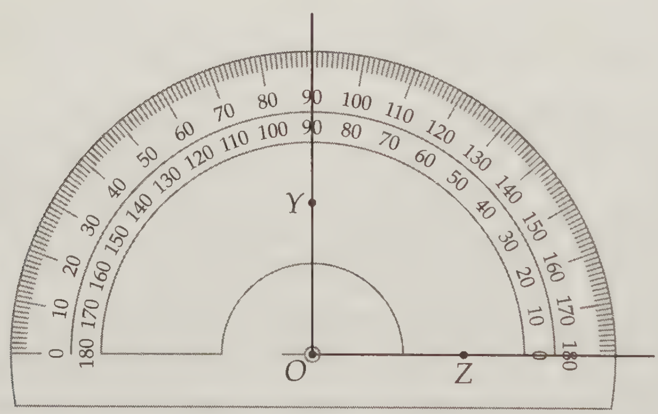
Problem 2.1: Use your protractor to find $\angle YOZ$, $\angle CXD$, $\angle DXE$, and $\angle CXE$.



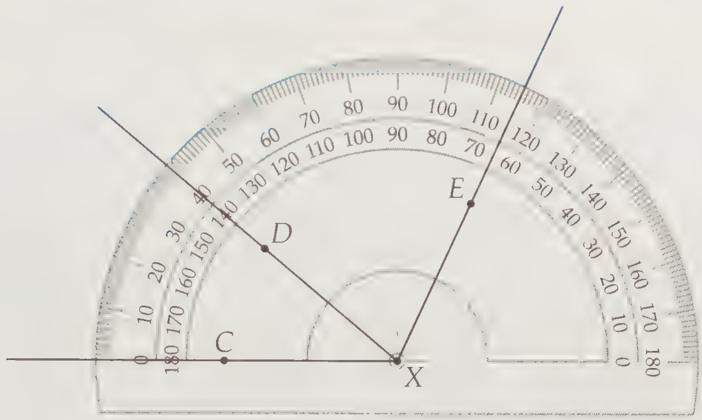
Solution for Problem 2.1: The protractor itself is half a circle (which we call a **semicircle**); we use it to measure the number of degrees of a circle the angle cuts off. Here are the steps we follow to use our protractor to measure angles:

1. Place the protractor on the angle so that the vertex of the angle is exactly where the center of the circle would be if the protractor were a whole circle. Your protractor should clearly show this center point; it's near the middle of the straight side.
2. Turn the protractor so that one side of the angle is along the 'zero line'; i.e., the line through the center point along the straight edge of the protractor.
3. Find where the other side of the angle meets the curved side of the protractor. The number there tells you the measure of the angle.

For $\angle YOZ$, we put our protractor on the page as shown below. We line up side $\overrightarrow{OZ}$ of the angle with the zero line of the protractor, placing the center point of the protractor over O. We find that side $\overrightarrow{OY}$ hits the curved edge at $90°$.

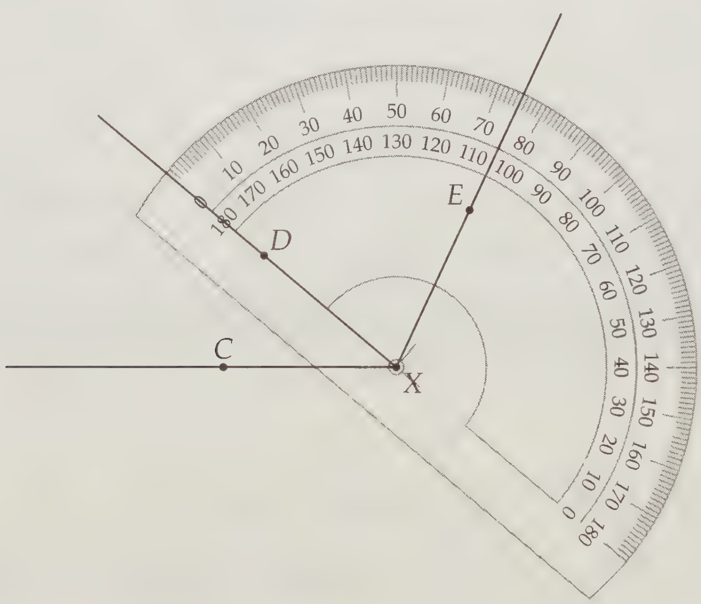


When we follow this procedure with $\angle CXD$, we find that there are two numbers where $\overrightarrow{XD}$ meets the curved edge in the following diagram. We know to take the smaller of these numbers – clearly there are 40 degrees, not 140 degrees, between $\overrightarrow{XC}$ and $\overrightarrow{XD}$. We can also note that $\angle CXD$ is less than half the entire semicircle, so its measure must be the smaller of the two numbers where $\overrightarrow{XD}$ meets the curved edge of the protractor.



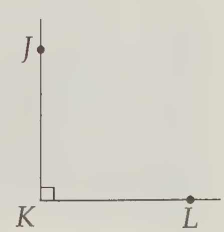
We can also use the above diagram to find the measure of $\angle CXE$. Once again, our angle hits a point on the curved edge with two numbers, but this time we know the angle is greater than 90° (since the angle is more than half the semicircle). Thus, we know that $\angle CXE = 115°$.

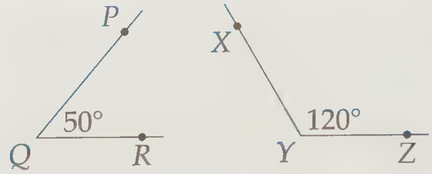
Finally, we can place the protractor as in the diagram below to find that $\angle DXE = 75°$.



Notice that $\angle CXD + \angle DXE = \angle CXE$. This isn't an accident! Since $\angle CXD$ and $\angle DXE$ share a side and a vertex, putting them together gives $\angle CXE$. □

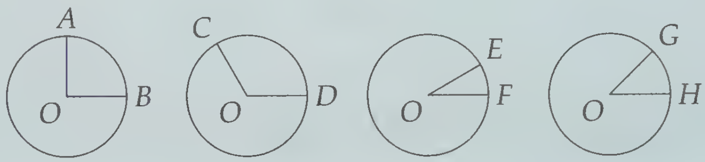
We saw in Problem 2.1 that knowing whether an angle is greater than or less than 90° is necessary for finding its measure using a protractor. This 90° is such an important measure that angles that are 90° have a special name, **right angles**. We usually mark right angles with a little box as shown in $\angle JKL$ at right. Two lines, rays, or line segments that form a right angle are said to be **perpendicular**. $\overline{JK}$ and $\overline{KL}$ are perpendicular; we can use the symbol $\perp$ to write this briefly: $\overline{JK} \perp \overline{KL}$.





Angles that are less than 90° are called **acute**, and those that are greater than 90° but less than 180° are called **obtuse**. Sometimes we write the measure of an angle inside the angle as shown above.

Problem 2.2: The diagram below shows four common angles. In each case, point O is the center of the circle. $\angle AOB$ cuts off 1/4 of a circle, $\angle COD$ cuts off 1/3 of a circle, $\angle EOF$ cuts off 1/12 of a circle, and $\angle GOH$ cuts off 1/8 of a circle.



(a) What is the measure in degrees of $\angle AOB$?

(b) What is the measure in degrees of $\angle COD$?

(c) What is the measure in degrees of $\angle EOF$?

(d) What is the measure in degrees of $\angle GOH$?

(e) Why in the world do we use such a weird number, 360, for the number of degrees in a whole circle?

Do not use a protractor; use what you are told about the angles in the text.

Solution for Problem 2.2: Since a whole circle is 360°, and $\angle AOB$ is 1/4 of a circle, we have

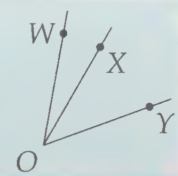
$$\angle AOB = \left(\tfrac{1}{4}\right)(360^\circ) = 90^\circ.$$

We can tackle the other three angles in exactly the same way:

$$\begin{aligned} \angle COD &= \left(\tfrac{1}{3}\right)(360^\circ) &&= 120^\circ \\ \angle EOF &= \left(\tfrac{1}{12}\right)(360^\circ) &&= 30^\circ \\ \angle GOH &= \left(\tfrac{1}{8}\right)(360^\circ) &&= 45^\circ \end{aligned}$$

The number 360 comes from the ancient Babylonians. The Babylonians used a number system with 60 digits, instead of our decimal system, which only has 10 digits. When choosing a number of degrees for a whole circle, they were likely influenced by their number system and possibly by astronomy (a year has around 360 days). However, a look at our answers above points to what might have been the largest factor in choosing 360. The Babylonians, like most people today, probably hated fractions. Since 360 is divisible by lots of different numbers, many common angles in geometry have integer measures. Had we used 100 degrees instead, we'd have to deal with measures such as $12\frac{1}{2}$ degrees, $8\frac{1}{3}$ degrees, and so on. □

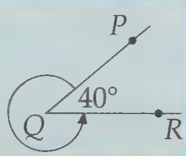
Problem 2.3: Given that $\angle WOY = 60°$ and $\angle WOX = 20°$, find $\angle XOY$.



Solution for Problem 2.3: As we just saw in Problem 2.1, we can combine two angles that share a side and a vertex to make a third whose measure equals the sum of the measures of the first two. Applying this to the diagram in the problem, we see that $\angle WOX + \angle XOY = \angle WOY$. Therefore, $20° + \angle XOY = 60°$, so $\angle XOY = 40°$. □

We call angles that share a side, like $\angle WOX$ and $\angle XOY$ in Problem 2.3, **adjacent angles**.

Problem 2.4: Suppose instead of measuring an angle the 'regular' way, we go the 'long' way around, as shown in the diagram. The 'regular' angle PQR has measure 40°. What is the measure of the 'long' way around angle?



Solution for Problem 2.4: If we imagine our 'regular' angle PQR cutting off a circle, we know it cuts off 40° of the circle. The 'long' way around then must be the rest of the circle. Since a whole circle is 360°, the remainder of our circle is $360° - 40° = 320°$. □

Angles that are greater than 180° are called **reflex angles**. They are rarely important in problems.

We finish our discussion of measuring angles by learning how to draw them given angle measurements.

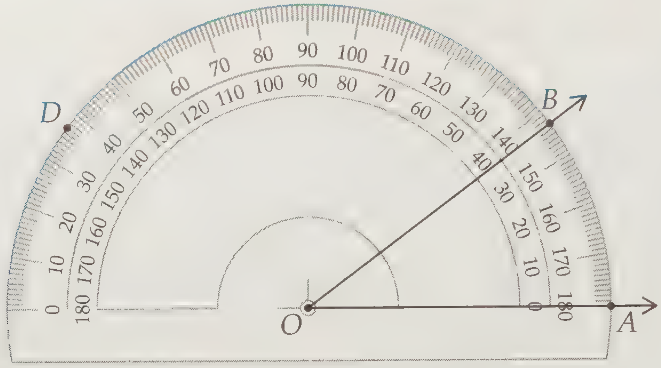
Problem 2.5: Use your protractor to create a 37° angle and a 143° angle.

Solution for Problem 2.5: We start with one side, $\overrightarrow{OA}$, of the 37° angle, which we can draw anywhere. To create the other side, we use our protractor to figure out where the other side would have to go in order to make a 37° angle. We place our protractor over $\overrightarrow{OA}$ as if we are measuring an angle with $\overrightarrow{OA}$ as a side. We then find the 37° point on the curved side, since a 37° angle would have to go through this point. As we've noticed before, our protractor has two 37's, one on each side. Since a 37° angle is clearly acute, we know to choose point B in the figure on the next page, thus making an angle that is less than 90°.

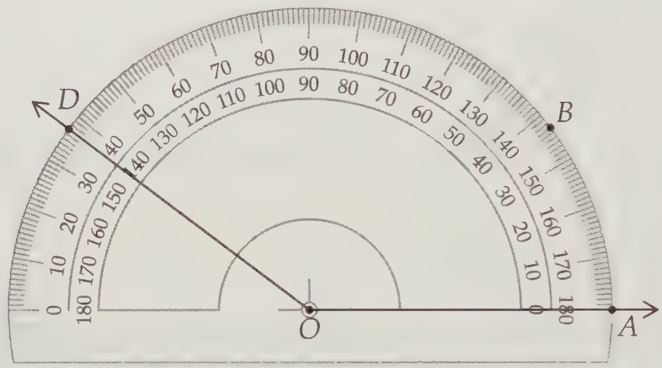
Extra! The golden ratio that we discussed on page 12 does not only appear in geometry! For example, consider the **Fibonacci sequence** shown below.

$$1, 1, 2, 3, 5, 8, 13, 21, 34, 55, 89, 144, 233, \ldots$$

The first two terms of the sequence are both 1, and each subsequent term is the sum of the previous two terms. Calculate the ratio between each term and the term before it, such as $34/21 \approx 1.619$. See anything interesting?



Similarly, we have two choices when we build our 143° angle. This time we choose the one that creates the obtuse angle, as shown in the diagram below.



□

Exercises

2.2.1 Use your protractor to make angles with the following measures:

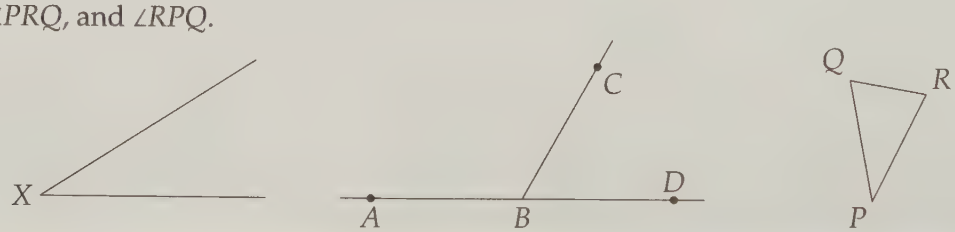
(a) 90° (b) 45° (c) 135° (d) 220°

2.2.2 Use your protractor to measure the angles shown. Classify each angle as right, acute, or obtuse.

(a) $\angle X$.

(b) $\angle ABC$ and $\angle DBC$.

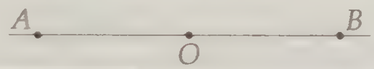
(c) $\angle PQR$, $\angle PRQ$, and $\angle RPQ$.



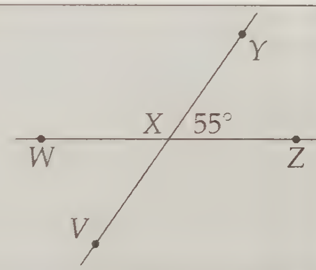
2.3 Straight and Vertical Angles

Problems

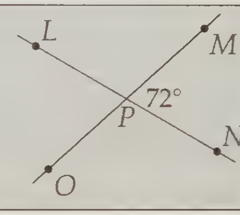
Problem 2.6: In the figure below, AOB is a straight line. What is the measure of $\angle AOB$?



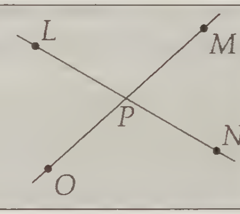
Problem 2.7: In the figure, lines $\overleftrightarrow{VY}$ and $\overleftrightarrow{WZ}$ meet at X, and $\angle YXZ = 55^\circ$. What is the measure of $\angle WXY$?



Problem 2.8: Lines $\overleftrightarrow{LN}$ and $\overleftrightarrow{MO}$ intersect at P such that $\angle MPN = 72^\circ$. Find $\angle LPO$.



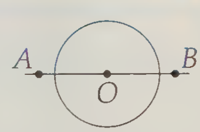
Problem 2.9: Lines $\overleftrightarrow{LN}$ and $\overleftrightarrow{MO}$ intersect at P. Prove that $\angle MPN = \angle LPO$.



As we have noted, two intersecting lines make angles. Can we make an angle with just one line?

Problem 2.6: In the figure, AOB is a straight line. What is the measure of $\angle AOB$?

Solution for Problem 2.6: If we don't see the answer right away, we can try to figure out what portion of a circle the angle cuts off. So, we draw a circle with center O as in the diagram to the right. Now we can see that the angle cuts off half a circle (whichever side of the line we pick). So, $\angle AOB = (1/2)(360^\circ) = 180^\circ$.

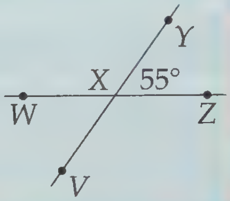


This one's easy to remember: we call an angle that is really a straight line a **straight angle**. □

Extra! *What science can there be more noble, more excellent, more useful for people, more admirably high and demonstrative, than this of mathematics?* –Benjamin Franklin

Straight angles appear too simple to be useful, but often the simplest tools are the best.

Problem 2.7: In the figure, lines $\overleftrightarrow{VY}$ and $\overleftrightarrow{WZ}$ meet at X and $\angle YXZ = 55°$. What is the measure of $\angle WXY$?

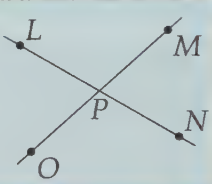


Solution for Problem 2.7: Since $\angle WXY$ and $\angle YXZ$ together make $\angle WXZ$, which is a straight angle, we know that $\angle WXY + \angle YXZ = 180°$. Hence, $\angle WXY = 180° - \angle YXZ = 125°$. □

We call two angles that add to 180° **supplementary angles**. As we have seen, when two lines intersect like $\overleftrightarrow{VY}$ and $\overleftrightarrow{WZ}$ in Problem 2.7, any two adjacent angles thus formed are supplementary because together they make a straight line.

Similarly, we call angles that add to 90° **complementary angles**.

Problem 2.8: Lines $\overleftrightarrow{LN}$ and $\overleftrightarrow{MO}$ intersect at P such that $\angle MPN = 72°$. Find $\angle LPO$.



Solution for Problem 2.8: Angle LPO sure looks equal to $\angle MPN$, and it 'makes sense' that the two are equal, but 'makes sense' isn't good enough in mathematics. We need proof, which we'll tackle in the next problem. For now, we'll try to compute $\angle LPO$.

Since it's not obvious how to compute $\angle LPO$, we start by finding angles we can measure. Since $\angle MPN$ and $\angle NPO$ together make a straight angle, we have $\angle NPO + \angle MPN = 180°$. Thus, $\angle NPO = 180° - 72° = 108°$.

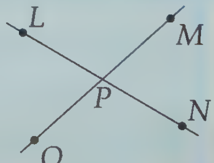
Similarly, since $\angle LPO$ and $\angle NPO$ are supplementary, we have

$$\angle LPO = 180° - \angle NPO = 180° - 108° = 72°.$$

□

Concept: When you can't find the answer right away, try finding whatever you can – you might discover something that leads to the answer! Better yet, you might learn something even more interesting than the answer. The best problem solvers are explorers.

Problem 2.9: Lines $\overleftrightarrow{LN}$ and $\overleftrightarrow{MO}$ intersect at P. Prove that $\angle MPN = \angle LPO$.



Solution for Problem 2.9: What's wrong with this proof:

Bogus Solution: Suppose $\angle MPN = 72^\circ$. Since $\angle MPO$ is a straight angle, we know that $\angle NPO = 180^\circ - 72^\circ = 108^\circ$. Similarly, we have $\angle LPO = 180^\circ - \angle NPO = 72^\circ$. Therefore, $\angle MPN = \angle LPO$.

Every statement in that 'proof' is true. However, it is not a complete proof because it only addresses one case; it only shows that $\angle MPN = \angle LPO$ when $\angle MPN = 72^\circ$.

WARNING!! An example is not a proof!

While examples aren't proofs, they can be useful as guides. Looking at our example, we can quickly construct our proof.

Since $\overleftrightarrow{MPO}$ is a line, we have

$$\angle MPN = 180^\circ - \angle NPO.$$

Since $\overleftrightarrow{LPN}$ is a line, we have

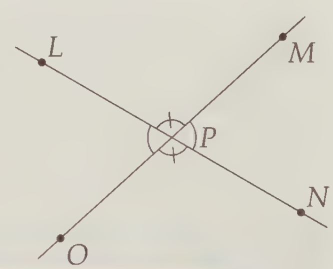
$$\angle LPO = 180^\circ - \angle NPO.$$

Combining these two equations gives $\angle MPN = 180^\circ - \angle NPO = \angle LPO$. □

Notice that our proof does not depend at all on the measure of $\angle MPN$. The proof works no matter how the lines intersect.

Pairs of angles such as $\angle MPN$ and $\angle LPO$ in the diagram below are called **vertical angles**. As we proved in Problem 2.9, vertical angles are always equal to each other.

We often use little arcs to mark equal angles. In the diagram to the right, $\angle MPN$ and $\angle LPO$ each have a single little arc in them to show that they are equal. Angles $\angle LPM$ and $\angle NPO$ also are vertical angles, so they are equal. We put a little hash-mark on the arcs at these angles to show that these two angles are equal to each other, but not necessarily equal to our first pair of equal angles (which have arcs without hash-marks).



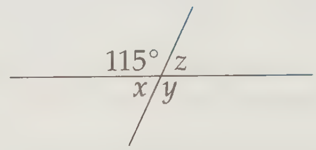
Important: Supplementary, right, obtuse, vertical, acute... by now the number of new names must be driving you nuts. Don't memorize what all these names mean now! The names are not that important. Besides, as you work your way through this book, you'll eventually see them so much you'll just know them anyway.

The concepts are more important than the words for solving problems. 'Angles like $\angle MPN$ and $\angle LPO$ in Problem 2.9 are equal' means something without any more information. 'Vertical angles are equal' doesn't tell you anything until you reach for your math dictionary to look up vertical angles.

The words, however, are important for communicating the concepts. For now, though, focus on the ideas. The words will come naturally.

Exercises

2.3.1 Find x, y, and z in the diagram below.



2.3.2 Find the measure of an angle that is supplementary to each of the following angles:

(a) $\angle AOB = 120^\circ$

(b) $\angle COD = 45^\circ$

(c) $\angle EOF = 90^\circ$

2.3.3 Find the measure of an angle that is complementary to each of the following angles:

(a) $\angle GOM = 30^\circ$

(b) $\angle IOJ = 45^\circ$

(c) $\angle KOL = 75^\circ$

2.4 Parallel Lines

Having dissected what happens when two lines meet, we should wonder about what happens if they don't. If two lines do not meet, we say that they are **parallel**. If lines $\overleftrightarrow{AB}$ and $\overleftrightarrow{CD}$ are parallel, we write $\overleftrightarrow{AB} \parallel \overleftrightarrow{CD}$.

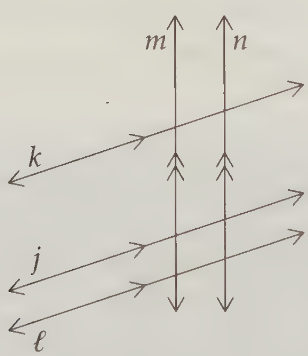


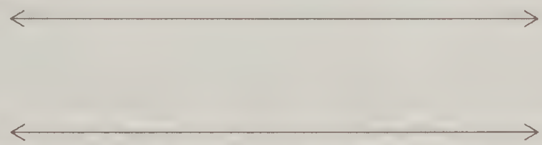
Figure 2.2: Two Sets of Parallel Lines

Just as we use little arcs to mark angles that are equal, we can use little arrows to mark lines that are parallel. In the diagram above, lines j, k, and ℓ are marked parallel, as are lines m and n. You won't see us use this notation all the time, though. Those little arrows can really clutter up a diagram.

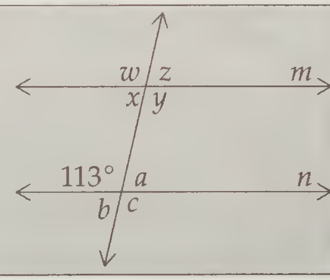
Problems

Problem 2.10: Draw a pair of parallel lines like those shown below. Then draw a line that crosses both of the parallel lines. Measure all the angles formed between your line and both of the parallel lines. Write the angle measures in the angles you form. Try it again with a different crossing line.

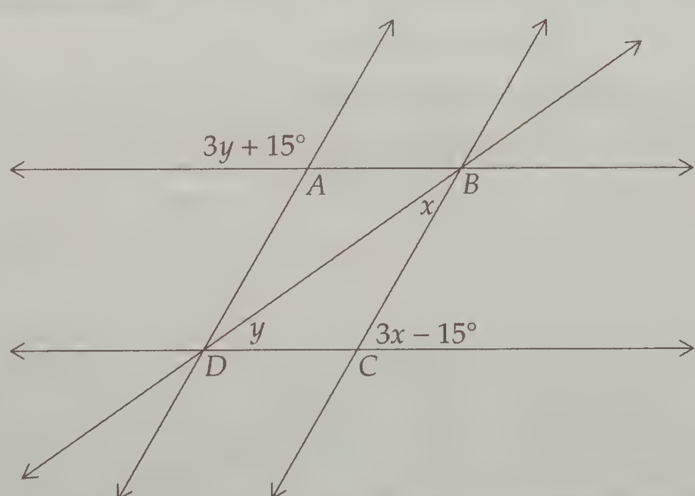
Do you notice anything interesting?



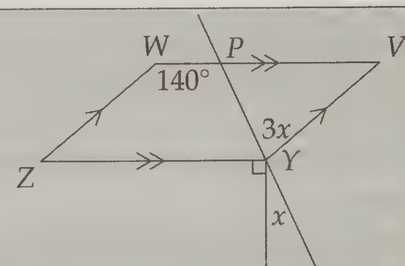
Problem 2.11: Lines m and n are parallel, and we are given the measure of one angle in the diagram as shown. Find the values of a, b, c, w, x, y, and z.



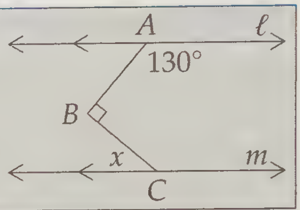
Problem 2.12: In the figure, we have $\overleftrightarrow{AB} \parallel \overleftrightarrow{CD}$ and $\overleftrightarrow{AD} \parallel \overleftrightarrow{BC}$. We are also given the measures of four angles as shown in terms of x and y. Find x and y.



Problem 2.13: Given that $\overline{WV} \parallel \overline{YZ}$ and $\overline{WZ} \parallel \overline{VY}$ in the diagram, find x.



Problem 2.14: In the diagram, $\ell \parallel m$ and the angles are as marked. Find x.

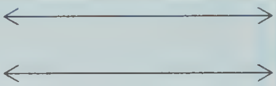


Back on page 10, we noted that one of our axioms (statements we must accept without proof) is

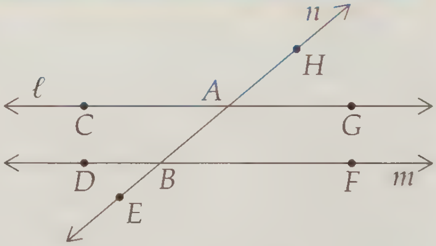
> Given any straight line and a point not on the line, there is exactly one straight line that passes through the point and never meets the first line.

This common-sense statement is sometimes called the **Parallel Postulate**. We'll start our exploration of parallel lines by taking a look at the angles formed when a line intersects a pair of parallel lines.

Problem 2.10: Draw a pair of parallel lines, then draw a line that crosses both of the parallel lines. Measure all the angles formed between your line and both of the parallel lines. Write the angle measures in the angles you form. Try it for a second line (don't worry about the angles where your two lines meet – just focus on the angles between the lines you draw and the parallel lines). Do you notice anything interesting?



Solution for Problem 2.10: In the diagram to the right, we have parallel lines ℓ and m, and we have added line n, which meets ℓ and m at A and B, respectively. We call a line that cuts across parallel lines a **transversal**. We measure $\angle HAG$ and find that it equals 40°. Since $\angle HAG$ and $\angle CAB$ are vertical angles, we don't even have to measure $\angle CAB$. We know that $\angle CAB = \angle HAG = 40^\circ$.



Since $\angle HAG$ and $\angle BAG$ together make up a straight angle, we don't have to measure $\angle BAG$. We know that $\angle BAG = 180^\circ - \angle HAG = 180^\circ - 40^\circ = 140^\circ$. Similarly, $\angle HAC = 140^\circ$.

We might wonder if we need our protractor at all, but then we think about those angles around B. They sure look equal to those around A, and common sense tells us that they are, but we measure to make sure. We find that indeed $\angle ABF = 40^\circ$, from which we deduce that $\angle DBE = 40^\circ$ as well. We can also quickly determine that $\angle ABD = 180^\circ - \angle ABF = 140^\circ$ and $\angle EBF = 140^\circ$.

Seeing that $\angle HAG = \angle ABF$, we wonder if it's always true that a transversal will cut parallel lines at equal angles like $\angle HAG$ and $\angle ABF$. Like the Parallel Postulate, this turns out to be one of those 'obvious' facts that cannot be proved. It must be assumed. As we have seen while finding the angles above, once we know that these two are equal, we can quickly use lines and vertical angles to find the rest of the angles.

Extra! *There is far more imagination in the head of Archimedes than in that of Homer.*

–Voltaire

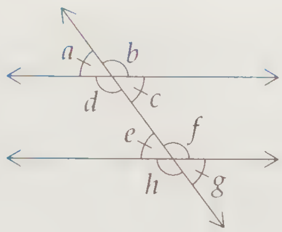


Figure 2.3: Angles Between a Transversal and Two Parallel Lines

Thus, we see that when a parallel line is cut by a transversal, we have two groups of four equal angles. Specifically, in Figure 2.3, we have

$$\begin{aligned} a &= c = e = g \\ b &= d = f = h \end{aligned}$$

Furthermore, the angles in the first group are supplementary to those in the second.

Pairs of these angles have special names to describe their relationships. These names are not terribly important, but you'll see them elsewhere.

a and e are **corresponding angles**.

d and f are **alternate interior angles**.

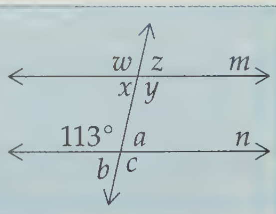
a and g are **alternate exterior angles**.

c and f are **same-side interior angles**.

b and g are **same-side exterior angles**.

Again, the names are not such a big deal. After doing enough geometry, you'll probably know them anyway. Don't bother memorizing them now. Just understand which angles are equal and which are supplementary. □

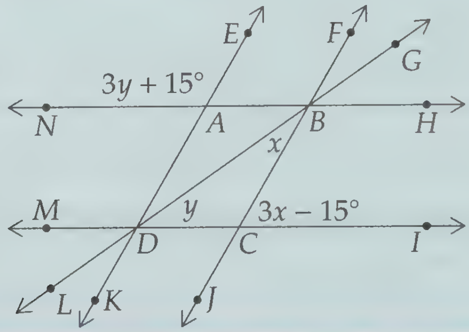
Problem 2.11: Lines m and n are parallel, and we are given the measure of one angle in the diagram as shown. Find the values of a, b, c, w, x, y, and z.



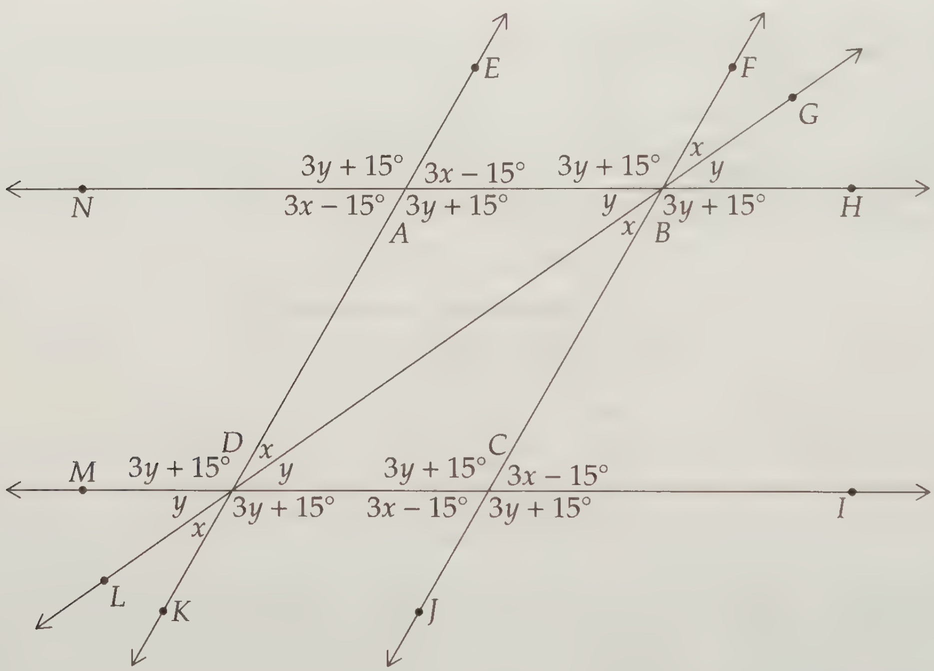
Solution for Problem 2.11: We know that when a transversal cuts parallel lines, equal angles come in groups of four as we saw in Problem 2.10. Therefore, we know that $w = y = c = 113°$. We also know that each angle in the other 'group of four' has a measure that is supplementary to 113°: $x = z = a = b = 180° - 113° = 67°$. □

Now that we understand the relationships between angles when a parallel line is cut by a transversal, let's try a more challenging problem.

Problem 2.12: In the figure, we have $\overleftrightarrow{AB} \parallel \overleftrightarrow{CD}$ and $\overleftrightarrow{AD} \parallel \overleftrightarrow{BC}$. We are also given the measures of four angles as shown in terms of x and y. Find x and y.



Solution for Problem 2.12: There's no obvious way to make an equation for x or y, so we start off by using our parallel lines and vertical angles to write the measures of all the angles we know in terms of x and y.



After labeling the angles we know in terms of x and y, we look for ways to build equations. We can use angles that together form straight angles at A and B:

$$\angle EAN + \angle EAB = (3y + 15^\circ) + (3x - 15^\circ) = 180^\circ$$

$$\angle FBA + \angle FBG + \angle GBH = (3y + 15^\circ) + x + y = 180^\circ.$$

Rearranging these gives

$$\begin{aligned} x + y &= 60^\circ \\ x + 4y &= 165^\circ \end{aligned}$$

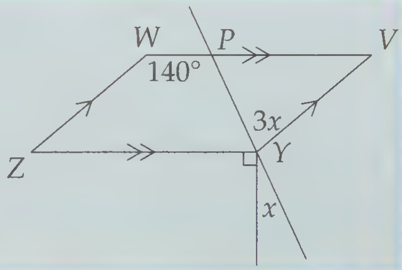
Subtracting the first from the second gives us $3y = 105^\circ$, so $y = 35^\circ$. We can then use substitution to find $x = 25^\circ$. Of course, we didn't have to label every angle above – we could have stopped when we had enough information to set up a pair of equations to solve for x and y.

Note that we could have used parallel line relationships to set up the equations, too. Since $\overleftrightarrow{AB} \parallel \overleftrightarrow{CD}$, we have $\angle DAN = \angle KDM$, so $3x - 15^\circ = x + y$. Also, $\angle HBC$ and $\angle BCI$ are supplementary, so $(3y + 15^\circ) + (3x - 15^\circ) = 180^\circ$. Solving these two equations gives us the same answer as before. (It better!) □

Concept: Solving a problem with two different methods is an excellent way to check your answer.

We'll finish with two more challenging problems that illustrate how useful parallel lines can be when seeking angle measures.

Problem 2.13: Given that $\overline{WV} \parallel \overline{YZ}$ and $\overline{WZ} \parallel \overline{VY}$ in the diagram, find x.



Solution for Problem 2.13: We start by using what we know about parallel lines to find as many angles as we can. We find $\angle V = 180^\circ - 140^\circ = 40^\circ$ since $\overline{WZ} \parallel \overline{VY}$. Similarly, $\angle Z = 40^\circ$, and $\angle ZYV = 180^\circ - \angle Z = 140^\circ$.

Since $\angle PYV = 3x$, we have $\angle PYZ = \angle VYZ - \angle PYV = 140^\circ - 3x$. Since this angle, the 90° angle, and the angle with measure x together give us straight line $\overleftrightarrow{PY}$, we have

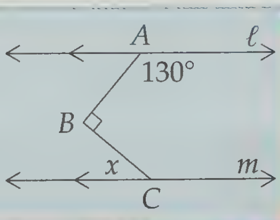
$$\angle PYZ + 90^\circ + x = 180^\circ.$$

We then substitute $\angle PYZ = 140^\circ - 3x$ into this equation, and we have $140^\circ - 3x + 90^\circ + x = 180^\circ$. We solve this equation for x to find that $x = 25^\circ$. □

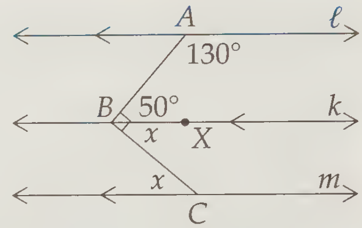
Using information about angles to find information about other angles is often called **angle-chasing**. We've already learned three important tools in angle-chasing: straight angles, vertical angles, and parallel lines. Stay tuned. We'll see plenty more!

Concept: Often when we're angle-chasing, our goal is to build an equation to solve for one of the variables in our problem.

Problem 2.14: In the diagram, $\ell \parallel m$ and the angles are as marked. Find x.



Solution for Problem 2.14: We need to relate our desired angle to angles we know, but neither $\overline{AB}$ nor $\overline{BC}$ cuts both parallel lines. However, if we add a line k through B parallel to ℓ and m, we can do some angle-chasing. Since $k \parallel \ell$, we have $\angle ABX = 180^\circ - 130^\circ = 50^\circ$. Since $k \parallel m$, we have $\angle XBC = x$. Since $\angle XBC + \angle ABX = \angle ABC$, we can now write an equation for x:



$$x + 50^\circ = 90^\circ.$$

Therefore, $x = 40^\circ$. □

> **Concept:** Parallel lines are so useful in problems involving angles that sometimes we'll add new ones to a diagram to help us.

Exercises

2.4.1 Find x and y in the diagram at left below.

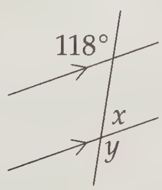


Figure 2.4: Diagram for Problem 2.4.1

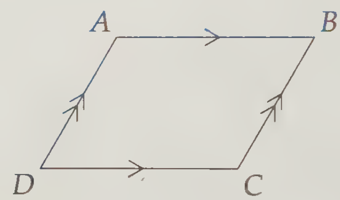


Figure 2.5: Diagram for Problem 2.4.2

2.4.2 Show that $\angle A = \angle C$ and $\angle B = \angle D$ in the diagram at right above.

2.4.3 Find x in the diagram at left below.

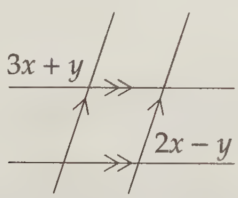


Figure 2.6: Diagram for Problem 2.4.3

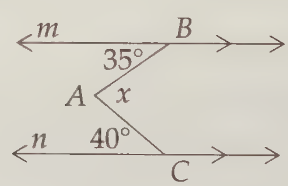
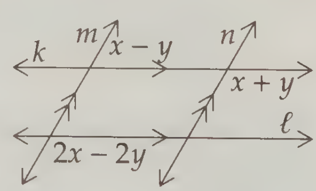


Figure 2.7: Diagram for Problem 2.4.4

2.4.4 In the diagram at right above, $m \parallel n$, and the angles are as marked. Find x. **Hints:** 224

2.4.5 In the diagram at right, $k \parallel \ell$ and $m \parallel n$. If the angles are as marked, find x and y. **Hints:** 481



> **Extra!** *Everything should be made as simple as possible, but not simpler.*
>
> –Albert Einstein

2.5 Angles in a Triangle

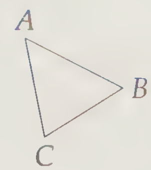


Figure 2.8: A Triangle

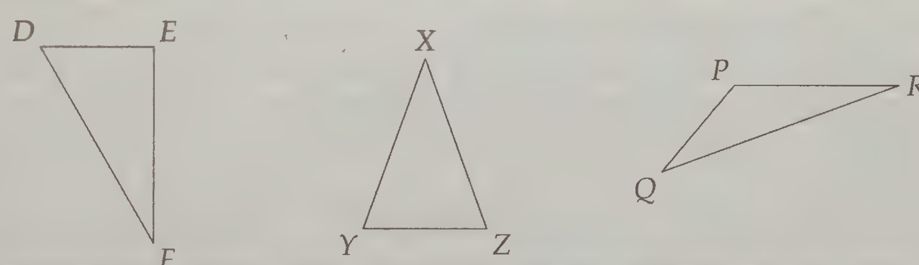
When we connect three points with line segments, we form a **triangle**, as shown in Figure 2.8 above. We will often refer to the triangle as $\triangle ABC$, or sometimes just ABC. The points A, B, and C are called **vertices** of the triangle, and the segments $\overline{AB}$, $\overline{BC}$, and $\overline{AC}$ are called **sides**. In this section, however, we will investigate the angles of a triangle.

Problems

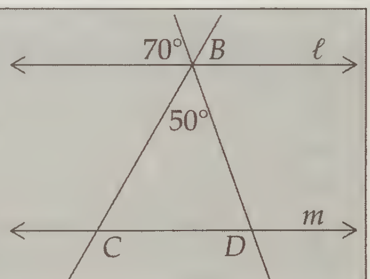
Problem 2.15:

(a) Find the measure of the three angles in each of the triangles below.

(b) Can you guess a statement that is always true about the sum of the angles in a triangle?



Problem 2.16: Given that $\ell \parallel m$, find $\angle BCD$ and $\angle BDC$.



Problem 2.17: Prove that the sum of the measures of the angles in a triangle is always 180°.

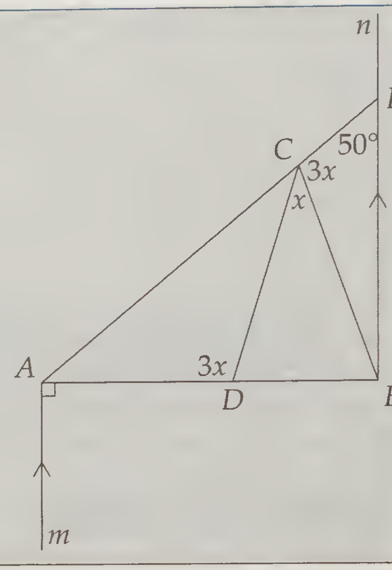
(a) Draw a triangle $\triangle ABC$, and a line k through point A such that $k \parallel \overline{BC}$.

(b) Find angles in your diagram that are equal to $\angle B$ and $\angle C$. Use little arcs as described on page 23 to mark the angles equal.

(c) Prove that $\angle CAB + \angle ABC + \angle BCA = 180°$.

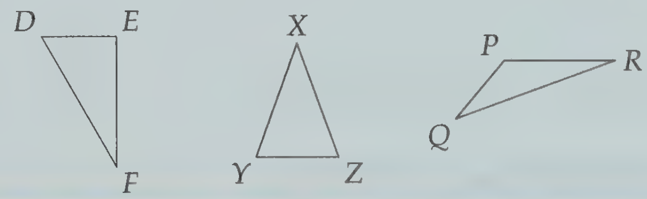
Problem 2.18: One angle in a triangle is twice another angle, and the third angle is 54°. What is the measure of the smallest angle in the triangle?

Problem 2.19: In the diagram, $m \parallel n$, $\overline{AB} \perp m$, $\angle ADC = \angle BCE = 3x$, $\angle CEB = 50°$, and $\angle BCD = x$. Find x.



We start our investigation of angles in a triangle by measuring angles in a few triangles.

Problem 2.15:
(a) Find the measure of the three angles in each of the triangles below.
(b) Can you guess a statement that is always true about the sum of the angles in a triangle?



Solution for Problem 2.15: Using our protractor, we can find the measures of the angles in each triangle as below:

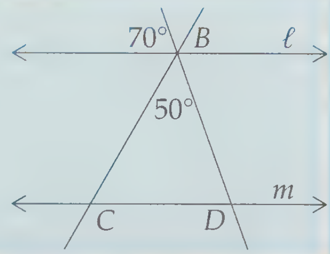
$$\begin{array}{llll} \triangle DEF: & \angle D = 60° & \angle E = 90° & \angle F = 30° \\ \triangle XYZ: & \angle X = 40° & \angle Y = 70° & \angle Z = 70° \\ \triangle PQR: & \angle P = 130° & \angle Q = 30° & \angle R = 20° \end{array}$$

In each case, we see that the sum of the angles is 180°. Hmmm... Is that always true? □

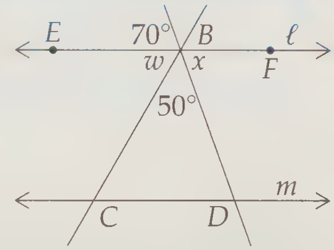
Before tackling the question of whether or not the angles of a triangle always add to 180°, let's try a parallel line problem that includes a triangle.

Extra! *The cowboys have a way of trussing up a steer or a pugnacious bronco which fixes the brute so that it can neither move nor think. This is the hog-tie, and it is what Euclid did to geometry.*
–Eric Temple Bell

Problem 2.16: Given that $\ell \parallel m$, find $\angle BCD$ and $\angle BDC$.



Solution for Problem 2.16: We start as we usually do when the answer isn't immediately obvious – we find what we can. We can use vertical angles to find $x = 70°$ in the figure to the right. We can also use either line ℓ or $\overleftrightarrow{BD}$ to find w. $\overleftrightarrow{BD}$ gives $70° + w + 50° = 180°$, so $w = 180° - 70° - 50° = 60°$.



Now we can use our parallel lines to find the angles we want. Since $\ell \parallel m$, we have $\angle BCD = w = 60°$ and $\angle BDC = x = 70°$. □

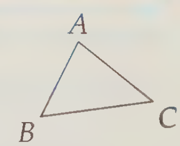
Notice that we don't have to call $\angle BCD$ and $\angle CBE$ 'alternate interior angles' in order to say they are equal. Once we state that $\ell \parallel m$, we can note that $\angle BCD = w$ without writing 'alternate interior angles.' However, if your teacher tells you that you have to include 'alternate interior angles,' you better do it!

Inspired by Problem 2.16, we can now prove that the sum of the angles in a triangle is always 180°.

Problem 2.17: Prove that the sum of the measures of the angles in a triangle is always 180°.

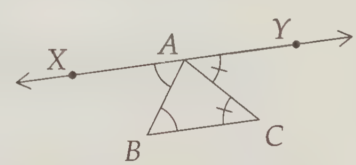
Solution for Problem 2.17: We start by drawing a triangle and by writing what we want to prove:

$$\angle ABC + \angle CAB + \angle BCA = 180°.$$



We don't know a whole lot about angles yet, but we can start by wondering '*Where have we seen* 180° *before?*' Answer: A straight angle. So, we'd like to find a straight line that has all three of our angles, just like $\angle EBC$, $\angle CBD$, and $\angle DBF$ make up line ℓ in our solution to Problem 2.16. However, we don't have any such line yet, so we'll have to add a line somewhere.

We need a line that allows us to use what we know about angles. We don't know much about angles, so we don't have too many options to investigate. So far, probably the most useful angle information we have learned comes from parallel lines. So, we add a line through A parallel to $\overline{BC}$ to create the diagram to the right, which looks curiously like the figure from Problem 2.16.



Since $\overleftrightarrow{XY} \parallel \overline{BC}$, we have $\angle XAB = \angle ABC$ and $\angle BCA = \angle CAY$. Since $\angle XAY$ is a straight angle, we have

$$\angle XAB + \angle CAB + \angle CAY = 180°.$$

Now we're home! We can now use the equalities we found with the parallel lines to get:

$$\angle ABC + \angle CAB + \angle BCA = 180°.$$

□

Important: The sum of the angles in a triangle is always 180°.

There are few geometric relationships you will use more than this one!

Important: Don't view the proof for Problem 2.17 as magic. We see 180°, and that makes us think of finding useful lines. We need a line that has angles equal to the three in $\triangle ABC$. Equal angles make us think of parallel lines, which always give lots of equal angles when cut by a transversal line.

Close the book and try to re-create this proof on your own!

Now let's try solving a problem using the fact that the sum of the angles of a triangle is 180°.

Problem 2.18: One angle in a triangle is twice another angle, and the third angle is 54°. What is the measure of the smallest angle in the triangle?

Solution for Problem 2.18: We know one angle is 54°, but all we know about the other two is that one is twice the other. Therefore, we let the smaller angle be x, so the larger is $2x$. Since the sum of the angles of a triangle is 180°, we have

$$54^\circ + x + 2x = 180^\circ.$$

Solving this, we find $x = 42^\circ$. The angles of our triangle are 42°, 54°, and 84°. (Note we can make a quick check by adding the three and making sure we get 180°.)

Our smallest angle is 42°. □

Concept: The key to tackling word problems in geometry is the same as any other kind of word problem – turn the words into math. Usually this means defining variables and using the words to write equations to solve for the variables.

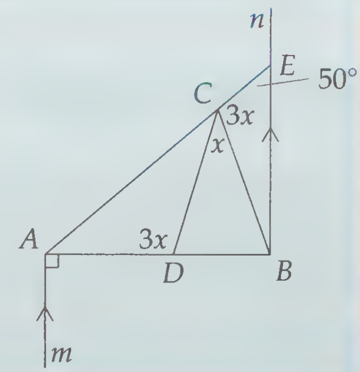
WARNING!! Your last step should be to make sure you've answered the question that is asked.

One of the most common uses of the fact that the angles in a triangle add up to 180° is as an angle-chasing tool. Let's give it a try.

Extra! *In the space of one hundred and seventy-six years the Mississippi has shortened itself two hundred and forty-two miles. Therefore, in the Old Silurian Period the Mississippi River was upward of one million three hundred thousand miles long. Seven hundred and forty-two years from now the Mississippi will be only a mile and three-quarters long. There is something fascinating about science. One gets such wholesome returns of conjecture out of such a trifling investment of fact.*

–Mark Twain

Problem 2.19: In the diagram, $m \parallel n$, $\overline{AB} \perp m$, $\angle ADC = \angle BCE = 3x$, $\angle CEB = 50°$, and $\angle BCD = x$. Find x.



Solution for Problem 2.19: We start with our parallel lines, and see that $\angle ABE = 90°$ because $m \parallel n$ and $m \perp \overline{AB}$. Then, from $\triangle ABE$ we have $\angle EAB + \angle ABE + \angle AEB = 180°$, so

$$\angle EAB = 180° - \angle ABE - \angle AEB = 40°.$$

We can then use $\triangle ACD$ to find

$$\angle ACD = 180° - \angle DAC - \angle ADC = 180° - 40° - 3x = 140° - 3x.$$

Concept: When angle-chasing, it's best to write the values you find for angles on your diagram as you find them, even when these values include variables.

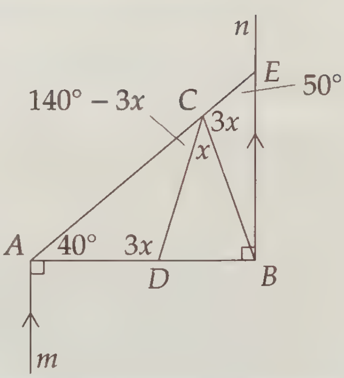
Our diagram now looks like the figure to the right. This picture suggests a way to finish the problem. We have three angles with vertices at C that together make a straight line, so we have

$$\angle ACD + \angle DCB + \angle BCE = 180°.$$

Substitution gives

$$140° - 3x + x + 3x = 180°,$$

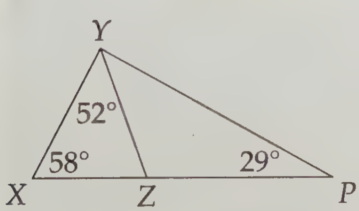
so $x = 40°$. □



Exercises

2.5.1 Two angles in a triangle have measures 30° and 57°. What is the measure of the third angle?

2.5.2 The angles in a triangle are in the ratio 1 : 2 : 3. What are the measures of the angles?



2.5.3 Find $\angle ZYP$ in the diagram at the left.

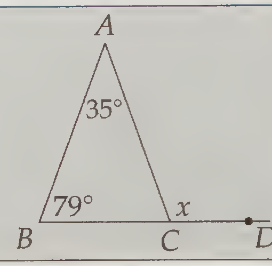
2.5.4 One of the angles in a triangle is a right angle. Show that the other two angles are complementary.

2.5.5★ Using what you know about triangles, find a different solution to Problem 2.14. **Hints:** 109

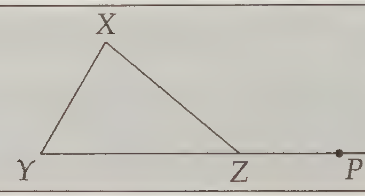
2.6 Exterior Angles

Problems

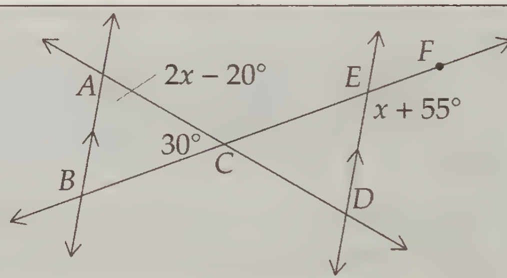
Problem 2.20: Find x given the angle measures shown.



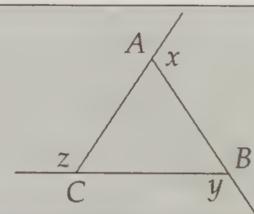
Problem 2.21: Prove that $\angle X + \angle Y = \angle XZP$ in the diagram shown.



Problem 2.22: In the diagram, $\overleftrightarrow{AB} \parallel \overleftrightarrow{DE}$, $\angle BAC = 2x - 20^\circ$, $\angle ACB = 30^\circ$, and $\angle DEF = x + 55^\circ$ as shown. Find $\angle CED$.

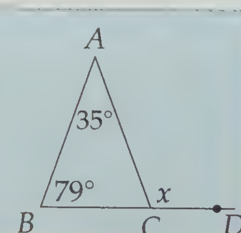


Problem 2.23: Find $x + y + z$.



In the previous section, we examined the **interior angles** of a triangle. In this section, we will take a look at the angles formed when we extend a side of a triangle past a vertex. We cleverly call these the **exterior angles** of the triangle. For example, in the problem below, $\angle ACD$ is an exterior angle of $\triangle ABC$.

Problem 2.20: Find x given the angle measures shown.



Solution for Problem 2.20: From $\triangle ABC$, we have

$$35^\circ + 79^\circ + \angle ACB = 180^\circ,$$

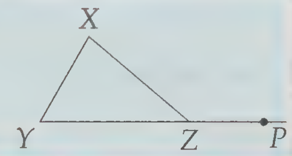
so $\angle ACB = 66^\circ$. From line $\overleftrightarrow{BD}$ we have

$$\angle ACB + x = 180^\circ,$$

so $x = 180^\circ - 66^\circ = 114^\circ$. □

That straightforward solution suggests we can prove something general about an exterior angle of a triangle.

Problem 2.21: Prove that $\angle X + \angle Y = \angle XZP$ in the diagram shown.



Solution for Problem 2.21: Our solution to Problem 2.20 guides the way. From $\triangle XYZ$, we have

$$\angle X + \angle Y + \angle XZY = 180^\circ.$$

From line $\overleftrightarrow{YZP}$, we have

$$\angle XZY + \angle XZP = 180^\circ.$$

Subtracting the second equation from the first gives us

$$\angle X + \angle Y - \angle XZP = 0,$$

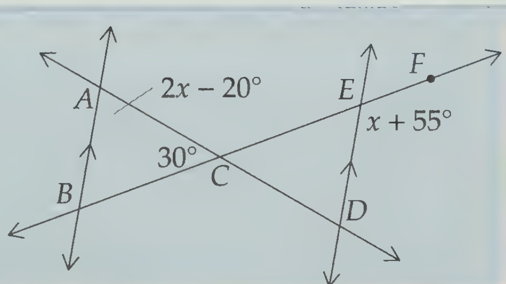
which we can easily rearrange to the desired $\angle X + \angle Y = \angle XZP$.

Notice that our proof does not at all depend on the values of the angles! □

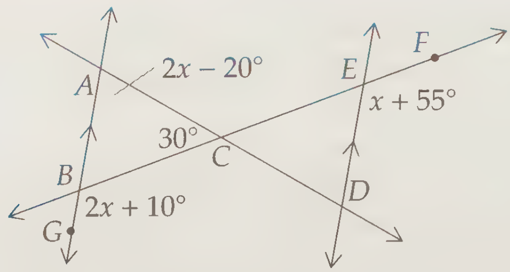
And now, we have more things to name. $\angle X$ and $\angle Y$ are called the **remote interior angles** of exterior angle $\angle XZP$ of $\triangle XYZ$. This name is *really* not important. We mostly give it a name so we can briefly write what we just proved:

Important: Any exterior angle of a triangle is equal to the sum of its remote interior angles.

Problem 2.22: In the diagram, $\overleftrightarrow{AB} \parallel \overleftrightarrow{DE}$, $\angle BAC = 2x - 20^\circ$, $\angle ACB = 30^\circ$, and $\angle DEF = x + 55^\circ$ as shown. Find $\angle CED$.



Solution for Problem 2.22: We know we'll probably need to find x to answer the problem. We could label every angle we know in terms of x, but first we take a minute to look for a faster way to get x.



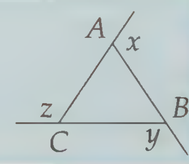
We have $\angle BAC$ and $\angle ACB$ of $\triangle ABC$, so we know the exterior angle $\angle GBC = \angle BAC + \angle ACB = 2x + 10°$. From $\overline{AB} \parallel \overline{DE}$, we know that $\angle GBC = \angle DEF = x + 55°$. Hence, we have $2x + 10° = x + 55°$, so $x = 45°$.

Our desired angle is the supplement of $\angle DEF$, so our answer is

$$\angle CED = 180° - \angle DEF = 180° - (x + 55°) = 80°.$$

There are many other ways we could have approached this problem. This is almost always true when we have problems involving exterior angles. Using exterior angles of a triangle is really just a shortcut for using a line and what we learned in Section 2.5 about the angles of a triangle. □

Problem 2.23: Find $x + y + z$.



Solution for Problem 2.23: We'll use $\angle A$, $\angle B$, and $\angle C$ to refer to the interior angles of $\triangle ABC$. Using what we just learned about exterior angles, we have

$$\begin{aligned} x &= \angle B + \angle C \\ y &= \angle A + \angle C \\ z &= \angle A + \angle B. \end{aligned}$$

Adding these, and noting that $\angle A + \angle B + \angle C = 180°$, gives:

$$\begin{aligned} x + y + z &= (\angle B + \angle C) + (\angle A + \angle C) + (\angle A + \angle B) \\ &= 2(\angle A + \angle B + \angle C) \\ &= 2(180°) \\ &= 360°. \end{aligned}$$

You'll be seeing this again. Only, next time, we'll be dealing with a figure that has more sides than a simple triangle! □

Extra! For an extra challenge, try to figure out the sum of the interior angles, and the sum of the exterior angles, in a figure with more sides than a triangle, such as the figure at right.

Exercises

2.6.1 Find y in the figure at left below.

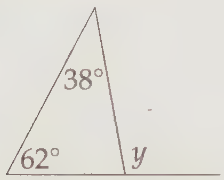


Figure 2.9: Diagram for Problem 2.6.1

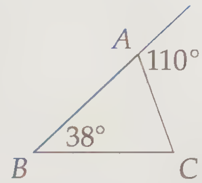
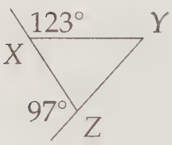


Figure 2.10: Diagram for Problem 2.6.2

2.6.2 Find $\angle C$ in the figure at right above.



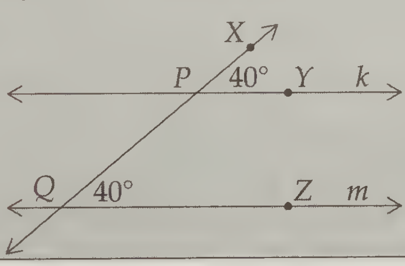
2.6.3 Find $\angle Y$ in the figure at left.

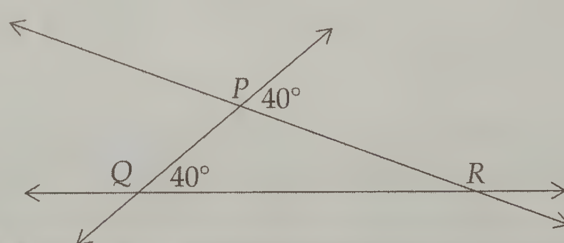
2.6.4 Must an exterior angle of a triangle always be greater than 90°?

2.7 Parallel Lines Revisited

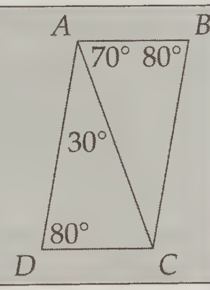
Problems

Problem 2.24:
(a) In the diagram at left below, what can we say about lines k and m?
(b) Why? (Hint: What's wrong with $\triangle PQR$ in the second figure?)



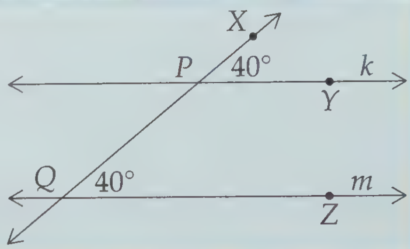


Problem 2.25: Given that the angles have the measures indicated in the diagram, prove that $\overline{AB} \parallel \overline{CD}$ and $\overline{BC} \parallel \overline{AD}$.



In Section 2.4, we learned some useful relationships about angles when we have parallel lines. In this section, we investigate whether or not we can 'go backwards' by using angle relationships to prove lines are parallel.

Problem 2.24: In the diagram to the right, what can we say about lines k and m? Why can we say it?



Solution for Problem 2.24: What's wrong with this solution:

Bogus Solution: If lines k and m were parallel, then $\angle XPY = \angle XQZ$. Since we know that $\angle XPY = \angle XQZ$, lines k and m must be parallel.

This 'solution' is exactly the same as saying the following:

If an animal is a cat, the animal must have four legs. My pet Spot has four legs. Spot must be a cat.

Clearly, this is bogus, because Spot could be a dog. All we know about Spot is that she has four legs. Our Bogus Solution to the problem has the same flaw!

We know for sure that if lines k and m were parallel, then $\angle XPY = \angle XQZ$. It is **not** logically valid to just run that backwards and say 'If $\angle XPY = \angle XQZ$, then lines k and m are parallel.' We have to prove this second statement separately. In Spot's case, we saw that the 'backwards' version of 'If an animal is a cat, then the animal must have four legs' is not even true! We call this 'backwards' version the **converse** of the original statement.

WARNING!! Suppose we have a true statement of the form:

If this, then that.

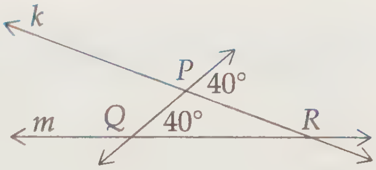
The converse of this statement is:

If that, then this.

Even if our original statement is true, the converse doesn't have to be true. We have to prove the converse separately.

And now, back to our story. We'd like to prove that lines k and m are parallel, and we can't use our angle relationships because we don't know they are parallel. The only other thing we know about parallel lines is that they never meet. So, we wonder if it is possible for the lines to meet if $\angle XPY = \angle XQZ$.

Suppose k and m meet at R as shown in the figure at right. We have a triangle, so we try to use what we know about triangles. First, we note that



$$\angle RPQ = 180^\circ - 40^\circ = 140^\circ.$$

Now that we have two angles of $\triangle PQR$, we can find the third:

$$\angle PRQ = 180^\circ - \angle RPQ - \angle RQP = 180^\circ - 140^\circ - 40^\circ = 0^\circ.$$

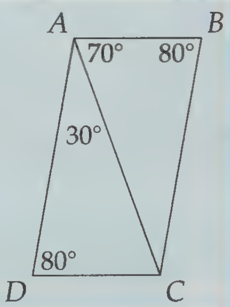
Uh-oh. If k and m meet at R, then $\angle R$ must be 0°, which doesn't make any sense. Therefore, we have shown that it is impossible for k and m to meet to the right of $\overleftrightarrow{PQ}$. (We'll leave the case of k and m meeting to the left of $\overleftrightarrow{PQ}$ as an Exercise.)

Since it is impossible for k and m to meet, they must be parallel. □

Important: Each of the angle relationships regarding parallel lines that we found in Section 2.4 can be used to prove that two lines are parallel. Using these angle relationships is the most common method of proving that two lines are parallel.

Concept: Our solution to the previous problem is an example of **proof by contradiction**. To prove a statement by contradiction, we start by assuming the statement is false. Then we show that this assumption leads us to an impossible statement (such as $\angle PRQ = 0^\circ$ above), which tells us that the assumption itself is false. Having proved the statement cannot be false, we have shown it must be true.

Problem 2.25: Given that the angles are as marked in the diagram, prove that $\overline{AB} \parallel \overline{CD}$ and $\overline{BC} \parallel \overline{AD}$.



Solution for Problem 2.25: We start by finding the angles we can find. Triangles $\triangle ACD$ and $\triangle ABC$ tell us that

$$\angle ACD = 180^\circ - 30^\circ - 80^\circ = 70^\circ \qquad \text{and} \qquad \angle ACB = 180^\circ - 70^\circ - 80^\circ = 30^\circ.$$

Therefore, we have $\angle DAC = \angle ACB$, which means that $\overline{AD} \parallel \overline{BC}$. We also have $\angle BAC = \angle DCA$, which tells us $\overline{AB} \parallel \overline{CD}$.

We also could have noted that $\angle D + \angle DAB = 180^\circ$, so $\overline{AB} \parallel \overline{CD}$. Likewise, $\angle B + \angle BAD = 180^\circ$ gives us $\overline{AD} \parallel \overline{BC}$. □

Exercises

2.7.1 We only did half of the proof in Problem 2.24. Complete the proof by showing that the two lines cannot meet to the left of $\overleftrightarrow{PQ}$.

2.7.2 For the diagram at left below, we have proved that if n cuts k and m such that $x = y$ (that is, corresponding angles are equal), then $k \parallel m$. Use this fact to show that if $y = z$ (that is, alternate interior angles are equal), then $k \parallel m$.

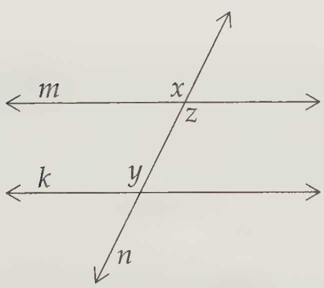


Figure 2.11: Diagram for Problem 2.7.2

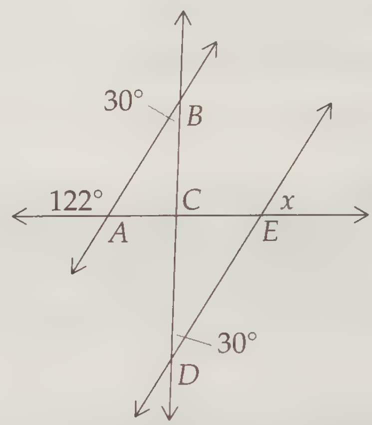


Figure 2.12: Diagram for Problem 2.7.3

2.7.3 Find x in the diagram to the right above. **Hints:** 427, 519

2.7.4 Write the converse of each of the following statements, then identify whether or not that converse is true.

(a) If two teams are playing in the World Cup Finals, then the teams must be playing soccer.

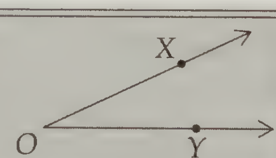
(b) If two of the angles of a triangle add to 80°, then one angle of the triangle must be 100°.

(c) If a river is the longest river in the world, then it must be the Nile.

(d) If an animal is a duck, then it must be a bird.

2.8 Summary

Definitions: Two rays that share an origin form an **angle**. The common origin of the rays is the **vertex** of the angle. We use the symbol $\angle$ to denote an angle, and we use a point on each side and the vertex, or just the vertex, to identify the angle, such as $\angle XOY$ at the right.



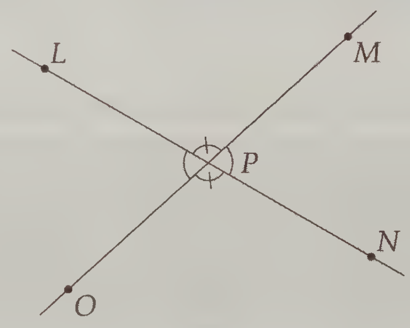
We can use a **protractor** to measure angles (see page 14). The **semicircular** arc of the protractor is divided into 180 degrees, so that a whole circle is 360 degrees.

Definitions: Angles can be classified by their measures.

- A 90° angle is a **right angle**. Lines, segments, or rays that form a right angle are said to be **perpendicular**.
- An angle smaller than 90° is an **acute angle**.
- An angle between 90° and 180° is an **obtuse angle**.
- An angle that measures 180° is a **straight angle**.
- An angle of more than 180° is a **reflex angle**.

Definitions:

- Two angles whose measures add to 180° are **supplementary angles**. Angles that together make up a straight angle form a particularly useful example of supplementary angles.
- Two angles whose measures add to 90° are **complementary angles**.
- When two lines intersect, they form two pairs of **vertical angles**, such as $\angle MPN$ and $\angle LPO$ below. Vertical angles are equal.



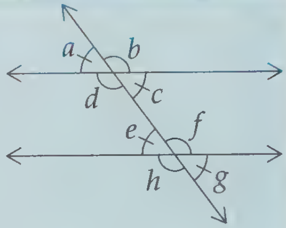
Angles $\angle LPM$ and $\angle NPM$ together form a straight angle, so they are supplementary (i.e. add to 180°).

Important: The concepts are more important than the words for solving problems. 'Angles like $\angle MPN$ and $\angle LPO$ above are equal' means something without any more information. 'Vertical angles are equal' doesn't tell you anything until you reach for your dictionary to look up vertical angles.

The words, however, are important for communicating the concepts. For now, though, focus on the ideas. The words will come naturally.

Definitions: Two lines that do not intersect are **parallel**. A line that cuts across multiple parallel lines is called a **transversal line**.

Important: The angles formed when a transversal cuts across two parallel lines come in two groups of four equal angles as shown:

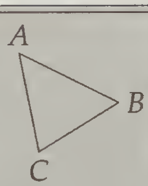


$$a = c = e = g$$
$$b = d = f = h$$

Each of the first set of angles is supplementary to each of the second set of angles. See page 27 for all the special names for pairs of these angles.

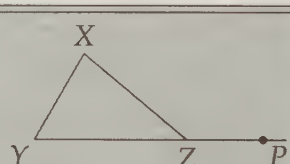
Important: The relationships described above when a transversal cuts two lines can also be used to show that two lines are parallel.

Definitions: When we connect three points with line segments, we form a **triangle**. The points are the **vertices** of the triangle, and the segments are the **sides** of the triangle. The angles inside the triangle formed by the sides are the **interior angles** of a triangle. When we refer to the angles of a triangle, we mean the interior angles.



Important: The sum of the angles in a triangle is always 180°.

Definitions: When we extend a side past a vertex of a triangle, we form an **exterior angle** of the triangle, such as $\angle XZP$ shown. We call $\angle X$ and $\angle Y$ the **remote interior angles** of exterior angle $\angle XZP$.



Important: Any exterior angle of a triangle is equal to the sum of its remote interior angles.

Problem Solving Strategies

Concepts:

- When you can't find the answer right away, try finding whatever you can – you might find something that leads to the answer! Better yet, you might find something even more interesting than the answer. The best problem solvers are explorers.
- Doing a problem two different ways is an excellent way to check your answer.

Continued on the next page. . .

Concepts: *. . . continued from the previous page*

- Often when we're angle-chasing our goal is to build an equation to solve for one of the variables in our problem.
- Parallel lines are so useful in problems involving angles that sometimes we'll add new ones to a diagram to help us.
- When angle-chasing, it's best to write the values you find for angles on your diagram as you find them, even when these values include variables.
- The key to tackling word problems in geometry is the same as any other kind of word problem – turn the words into math. Usually this means defining variables and using the words to write equations to solve for the variables.
- Sometimes using a **proof by contradiction** is much easier than proving a statement directly. To prove a statement by contradiction, we start by assuming the statement is false. Then we show that this assumption leads us to an impossible statement, which tells us that the assumption itself is false. Having proved the statement cannot be false, we have shown it must be true.

Things To Watch Out For!

WARNING!!

- An example is not a proof!
- Your last step should be to make sure you've answered the question that is asked.

Proof is at the heart of mathematics. On page 40, we saw one of the most common logical errors beginners make.

WARNING!! Suppose we have a true statement of the form

If this, then that.

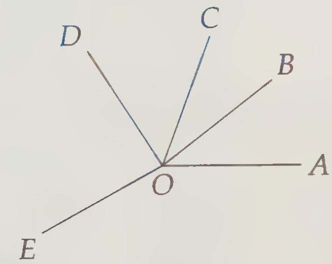
The **converse** of this statement is

If that, then this.

Even if our original statement is true, the converse doesn't have to be true. We have to prove the converse separately.

REVIEW PROBLEMS

2.26 Using your protractor, determine the following angles in the diagram to the right:



(a) $\angle AOB$

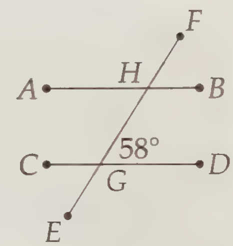
(b) $\angle AOC$

(c) $\angle BOC$

(d) $\angle DOE$

2.27 How many seconds does it take the second hand of a clock to rotate through an angle of $72°$?

2.28 Given $\overline{AB} \parallel \overline{CD}$ in the diagram to the right, find:



(a) $\angle CGE$

(b) $\angle HGC$

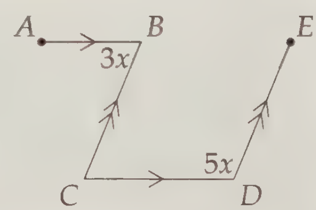
(c) $\angle FHB$

(d) $\angle BHG$

2.29 Two angles of a triangle are $30°$ and $70°$. What is the third angle?

2.30 The angle $\angle B$ in $\triangle ABC$ is $60°$. If an exterior angle at A is $170°$, what is $\angle C$?

2.31 Find x in the diagram to the right.



2.32 Lines $\overleftrightarrow{PQ}$ and $\overleftrightarrow{RS}$ are parallel, and $\overleftrightarrow{TV} \perp \overleftrightarrow{PQ}$. If $\overleftrightarrow{TV}$ intersects $\overleftrightarrow{PQ}$ at X and $\overleftrightarrow{RS}$ at Y, find $\angle RYX$.

2.33 In the diagram to the left below, the angles are as marked. Find x.

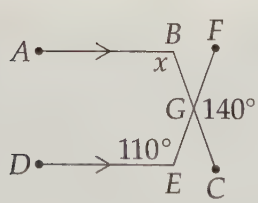


Figure 2.13: Diagram for Problem 2.33

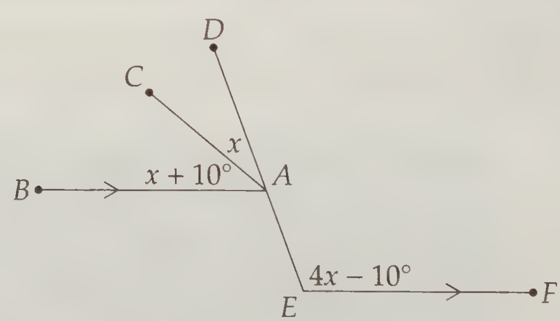


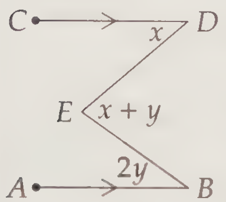
Figure 2.14: Diagram for Problem 2.34

2.34 The measures of the angles are as marked in the diagram to the right above. Find x.

2.35 The three angles $\angle A$, $\angle B$, and $\angle C$ have the property that $\angle A$ is complementary to $\angle B$, $\angle B$ is complementary to $\angle C$, and $\angle C$ is complementary to $\angle A$. Prove that $\angle A = \angle B = \angle C$.

2.36 Let $\triangle ABC$ have (interior) angles in the ratio 3 : 4 : 5. What is the measure of its smallest exterior angle? **Hints:** 135

2.37 The exterior angles of a triangle are in the ratio 2 : 3 : 4. What are the (interior) angles of the triangle?



2.38 Is it possible for the angles in the diagram to the left to have the measures indicated? Why or why not?

2.39 Is it possible for two exterior angles of a triangle to be supplementary? Why or why not? **Hints:** 491

2.40 Three straight lines intersect at O and $\angle COD = \angle DOE$ in the diagram at left below. The ratio of $\angle COB$ to $\angle BOF$ is 7 : 2. What is the number of degrees in $\angle COD$? *(Source: MATHCOUNTS)*

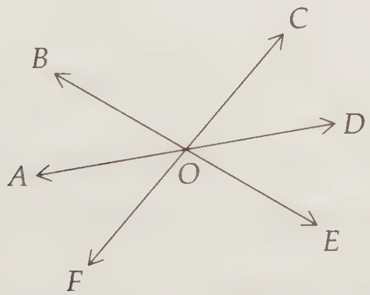


Figure 2.15: Diagram for Problem 2.40

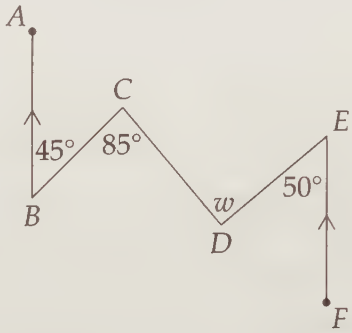
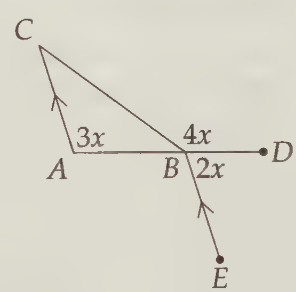


Figure 2.16: Diagram for Problem 2.41

2.41 Find w in the diagram to the right above. **Hints:** 248

2.42 One angle of a triangle is 20°. If the largest angle of the triangle has six times the measure of the smallest, what are the angles of the triangle?

2.43 The measures of the angles in the diagram to the right are as marked. Find $\angle C$.



2.44 The three angles of a triangle have measures $\angle A = x - 2y$, $\angle B = 3x + 5y$, and $\angle C = 5x - 3y$. Find x.

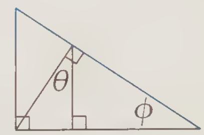
2.45 It is not possible to find the value of y in the previous problem from the given information. What if you are also told that one angle of the triangle is 10°? Is it now possible to compute y? What are the possible positive value(s) of y?

2.46 Show that if a transversal cuts two lines such that the same-side interior angles are supplementary, then the two lines are parallel.

Extra! *Music is the pleasure the human mind experiences from counting without being aware that it is counting.*

–Gottfried Wilhelm von Leibniz

Challenge Problems



2.47 Three angles in the diagram to the right are marked as right angles. If $\phi = 27°$, what is the value of θ? *(Source: Mandelbrot)*

2.48 What is the number of degrees of the angle formed by the minute and hour hands of a clock at 11:10 PM? *(Source: MATHCOUNTS)* **Hints:** 52, 356, 282

2.49 Find x in the diagram at left below. **Hints:** 355

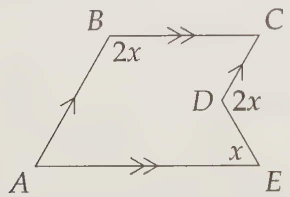


Figure 2.17: Diagram for Problem 2.49

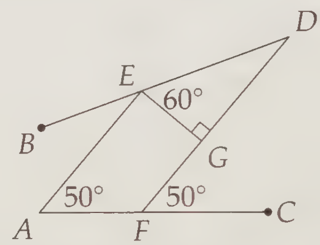


Figure 2.18: Diagram for Problem 2.50

2.50 The angles in the diagram to the right above are as marked. Find $\angle BEA$. **Hints:** 8

2.51 One angle of a triangle is equal to the sum of the other two. Show that the sum of two exterior angles of the triangle is 180° greater than the third. **Hints:** 151, 200

2.52 One of the angles of the triangle in the previous problem is equal to 40°. What are the measures of the other two angles?

2.53 The angles of a triangle are in arithmetic progression. If one of the angles is 100°, what are the measures of the other two angles? (An arithmetic progression is a sequence of numbers in which the difference between each term and its preceding term is always the same.) **Hints:** 581

2.54 It is possible for the interior angles of a triangle to be in the ratio 1 : 2 : 6, but is it possible for the *exterior* angles of a triangle to be in the ratio 1 : 2 : 6? Prove your answer. **Hints:** 14, 444

2.55 Find $\angle A + \angle B + \angle C + \angle D$ in the figure at left below. **Hints:** 358

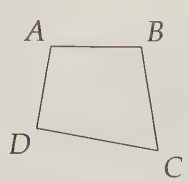


Figure 2.19: Diagram for Problem 2.55

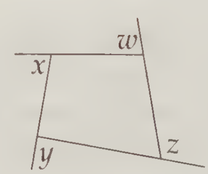


Figure 2.20: Diagram for Problem 2.56

2.56 Find $w + x + y + z$ in the figure at right above. **Hints:** 17, 400

2.57★ Point Z is on side $\overline{PR}$ of $\triangle PQR$ such that $\angle PZQ = \angle PQZ$, and $\angle PQR - \angle PRQ = 42°$. Find $\angle RQZ$. **Hints:** 54, 441, 156